

STUDIES IN MULTIPLE SCLEROSIS

VIII ETIOLOGIC FACTORS IN MULTIPLE SCLEROSIS*

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THE EXPERIMENTAL PRODUCTION OF LESIONS SIMULATING MULTIPLE SCLEROSIS

In previous communications, it has been pointed out that lesions closely resembling typical plaques of multiple sclerosis may be produced in animals in either of two ways—by the subcutaneous injection of minute doses of tetanus toxin¹ and by experimental obstruction of cerebral venules.² In each case sclerotic lesions could be demonstrated only after an interval of several months from the original injury. The earlier lesions closely resembled those of “post-infectious encephalomyelitis,” a disease occasionally seen following measles³ and vaccinia⁴ and often without obvious antecedent. Areas of demyelination, with relative persistence of axis cylinders, accompanied by infiltration in the early stages and gliosis later, have also been described as a result of carbon monoxide poisoning with venous thrombosis,⁵ and chronic cyanide poisoning.⁶

By themselves, these experimental results prove little in regard to the human disease of multiple sclerosis. They do suggest, however, that

1 “Post-infectious encephalomyelitis” and multiple sclerosis may represent the same pathologic process at different stages with variations in intensity.

2 Local anoxemia, or more specifically local venous obstruction, should receive consideration as the immediate cause of the primary demyelination.

3 The process may be precipitated in some way by an organic toxin, and probably by any one of many.

None of these propositions is entirely new, but apparently no attempt has been made to interrelate them. Indeed, they have been considered almost mutually exclusive. Thus multiple sclerosis has been looked upon as a toxic disease, and therefore not circulatory, or as an infectious disease and therefore not toxic. Before attempting to harmonize these points of view, let us see what may be said for each of them separately.

“ENCEPHALOMYELITIS” AND MULTIPLE SCLEROSIS AS MANIFESTATIONS OF THE SAME PATHOLOGIC PROCESS

The essential similarity between “encephalomyelitis” (especially the post-measles and post-vaccinal forms) and multiple sclerosis has frequently

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been affirmed,⁷ and almost as frequently denied⁸ The points of similarity between the two diseases which are usually pointed out are the patchy distribution of the lesions, which surround venules of the white matter, and the predominant involvement of myelin, axis cylinders often remaining relatively intact On the other side of the question, it has been asserted that the lesions of the acute disease are more diffuse than those of multiple sclerosis, and this is doubtless true in some instances In these cases the picture as a whole is much more similar to that of diffuse sclerosis or Schilder's disease This condition, however, is probably to be considered a variant of multiple sclerosis⁹ Further, it has been objected that there is much more extensive loss of axis cylinders in the lesions of "disseminated encephalomyelitis" than in typical sclerotic plaques This is probably true, but it must be remembered that the patient has died at the height of a severe acute process in most cases of the former disease coming to autopsy, while a milder process, such as multiple sclerosis, might be expected to allow the survival of axis cylinders in greater numbers Even in multiple sclerosis, extensive loss of axis cylinders is common, as all modern investigators have reported In a series of 35 separate plaques from 11 cases of multiple sclerosis studied with this question in mind, almost complete destruction of axis cylinders was found in eighteen¹⁰ A case reported by Walthard¹¹ is often cited as direct evidence that post-measles "encephalitis" is not the equivalent of multiple sclerosis The patient died three months after a typical onset, and at autopsy a considerable loss of axis cylinders was found in most of the lesions Perhaps more relevant, however, may be the cases of Cramer¹² and of Schlesinger,¹³ in both of which the patients survived an unmistakable post-infectious "encephalitis" for many months and in both of which typical sclerotic plaques (figure 1) containing relatively intact axones were found post mortem Walthard's case may then be considered a transitional form of which a number have been reported^{8, 14, 15}

Gliosis, which is a striking feature of multiple sclerosis, would scarcely be expected in an acute process It is, however, not as specific for multiple sclerosis as has sometimes been supposed Lesions with loss of myelin but with little or no gliosis occur side by side with typical sclerotic plaques, and on the other hand, areas of secondary degeneration in other pathologic conditions of comparable age and intensity with those of multiple sclerosis develop quite as dense a glial scar (Figures 2 and 3) It is probably a general rule that once myelin has been broken down, it cannot regenerate in the adult central nervous system,¹⁶ and that myelin loss is repaired by a glial fibrosis which grows denser and less cellular with the passage of years An exception to the rule is seen in such conditions as the degeneration accompanying primary anemia, in which gliosis appears to be inhibited by some nutritive lack

Additional points of similarity between "encephalomyelitis" and multiple sclerosis, which have apparently escaped notice, are the vascular ob-

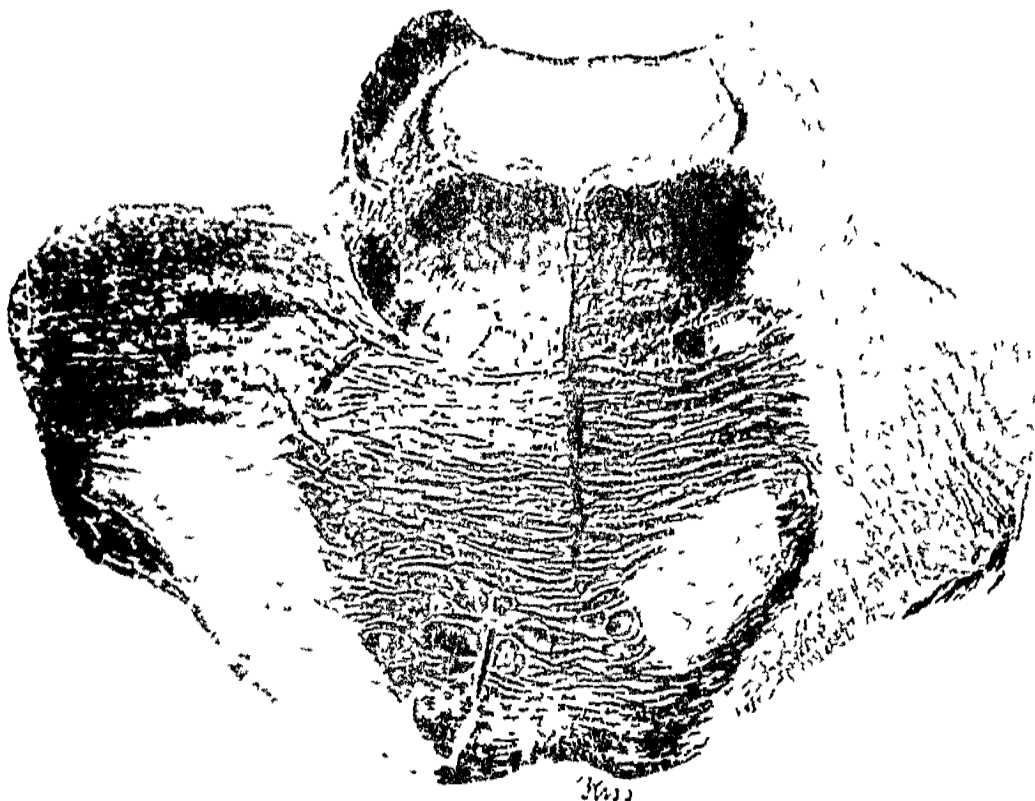


FIG 1 Typical sclerotic plaques in the pons, from the brain of a boy of 7, who died 22 months from the onset of a clinically typical post-measles "encephalomyelitis," after a course marked by remissions and exacerbations (After Schlesinger¹³)

structions and engorgement and the perivascular hemorrhages seen in both, as will be described below

It is assumed by most writers on the subject, often without any display of evidence, that an "encephalomyelitis" in which an infiltration with hematogenous cells occurs must be due to a filterable virus if it occurs in the course of a virus disease. Infiltration with lymphocytes and leukocytes is, however, common in the vicinity of softenings and tumors¹⁷. Further, Rivers¹⁸ has pointed out that all known virus diseases lead primarily to either a necrosis or a proliferation of parenchymal cells, or both, while in "encephalomyelitis" of the type in question the chief damage is to cell appendages, and in particular to the myelin which is probably to be considered as intercellular substance¹⁹.

VASCULAR LESIONS AND DEMYELINATION

In addition to the experimental data already presented, there is evidence from the pathology of other diseases that disturbance of circulation may lead to loss of myelin and to gliosis. Thus the ordinary forms of arterio-

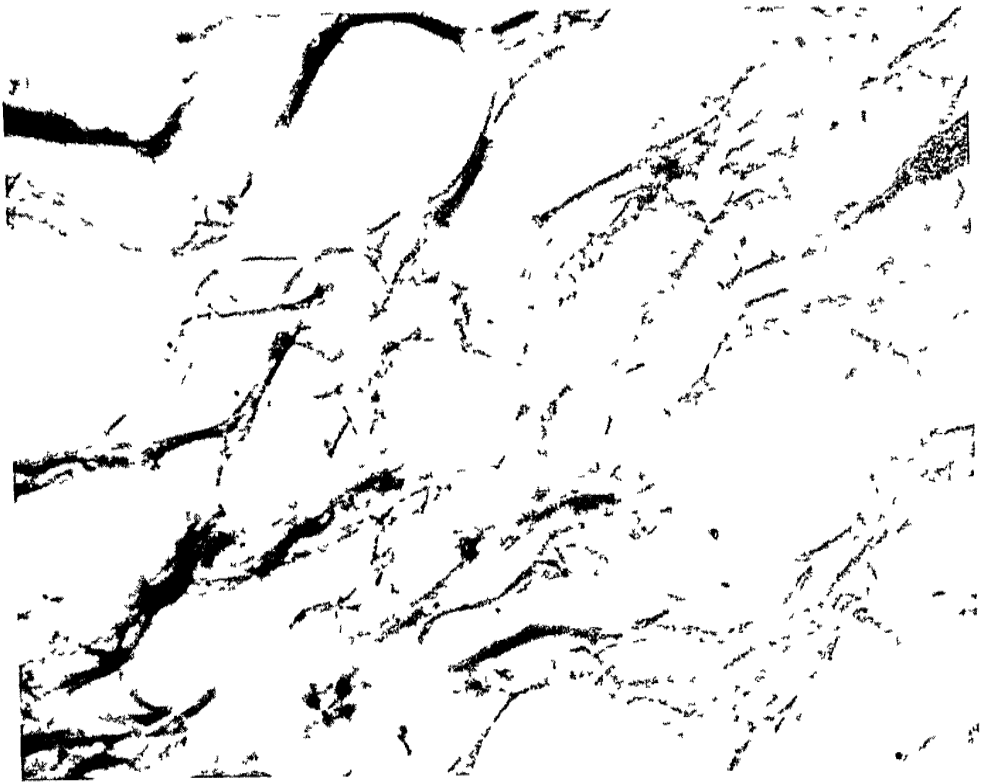
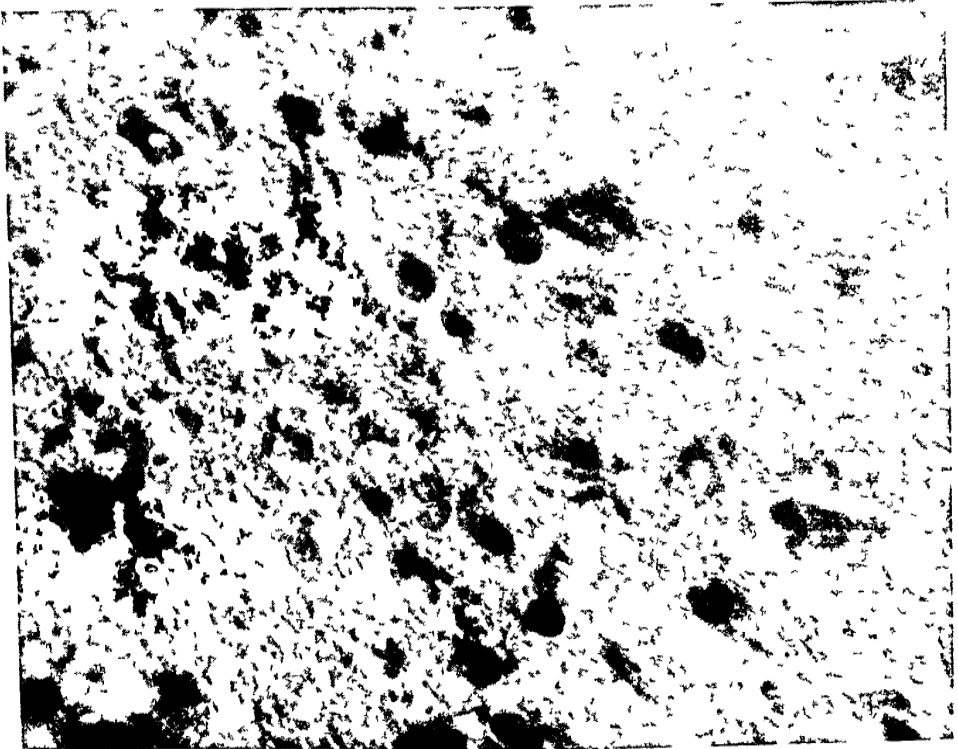


FIG 2



FIGS 2 and 3 Loose filmy gliosis (figure 2) from a plaque from a long-standing case of multiple sclerosis, for comparison with the dense gliosis in a degenerated pyramid (figure 3) secondary to a capsular apoplexy 10 years previous. Note also the absence of axis cylinders in the sclerotic plaque. Masson stain, oil immersion lens.

sclerosis may produce small lesions in the nervous system in which the myelin suffers far more than the axis cylinders. In human as well as in experimental carbon monoxide poisoning, areas of demyelination may be situated in relation to thrombosed veins,⁵ and in one well studied case, exposure to carbon monoxide apparently precipitated a fatal exacerbation of multiple sclerosis.²⁰ In the condition known as chronic subcortical encephalopathy, plaques closely resembling those of multiple sclerosis occur obviously as a result of vascular obstruction.²¹

Alterations in the blood vessels have long been recognized in multiple sclerosis, and formerly a good deal of emphasis was laid upon them as possibly of importance in the etiology of the disease, for example, by Williamson,²² Ribbert,²³ and Borst.²⁴ It was usually assumed that the vascular abnormality permitted the escape of a toxin or "toxic lymph" into the substance of the brain. More recent writers, such as Hassin,²⁵ have dismissed the vascular changes as secondary to the parenchymal degeneration. The perivascular infiltration may well be accepted as the cause of adventitial proliferation, but it seems open to question whether the obstructions, dilations, and perivascular hemorrhages which have been described could be explained on a similar basis.

In "post-infectious encephalomyelitis" vascular engorgement has been reported by most observers, thromboses by several, and perivascular hemorrhages by some. The presence of thrombi in post-measles "encephalitis" has received especial attention from Kreider²⁶ in Feriario's laboratory.

It seemed worth while, therefore, to make a special review of the subject. This proved not as easy as had been anticipated. Aside from the difficulties of collecting material,^{*} it was found that the ordinary routine stains and the usual neuropathologic methods were entirely inadequate for a study of thrombi and finer vascular changes. Finally, Masson's stain on paraffin material was tried, and appeared satisfactory. The results of the investigation will be published in detail elsewhere, and are merely summarized here.

In a case of post-vaccinal "encephalitis" and in another histologically similar case, but without known etiology, the vascular changes may be summarized as extreme engorgement, phagocytosis of brown pigment with and without perivascular hemorrhage and intravascular clot formation. In numerous vessels, large and small, there were masses of platelets, a heavy deposit of fibrin adherent to the walls and endothelial proliferation (figure 4).

In every one of 14 cases of multiple sclerosis in which material was available for adequate study, marked vascular changes were observed. The adventitial proliferation, doubtless a result of perivascular infiltration, has already been referred to. In addition, an engorgement, tortuosity and proliferation of veins in the vicinity of plaques were striking (figure 5).

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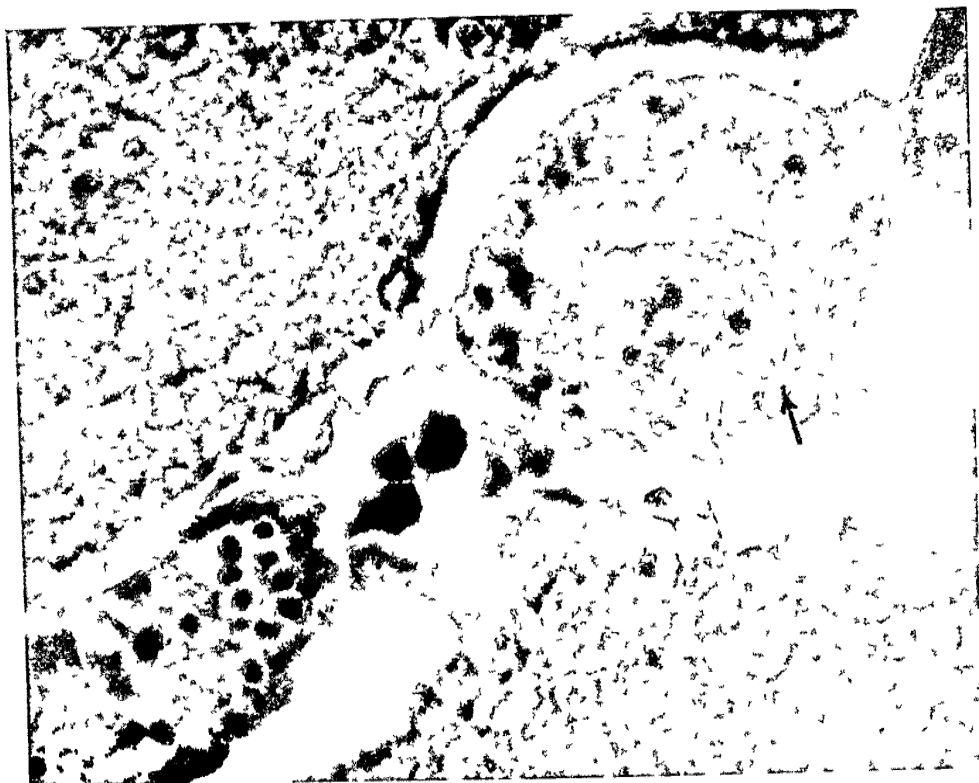


FIG 4 Post-vaccinal "encephalitis" Engorged vessel, containing a mass of fibrin adherent to the walls and clumps of leukocytes Cells are growing along a fibrin strand (arrow) Masson stain, oil immersion lens

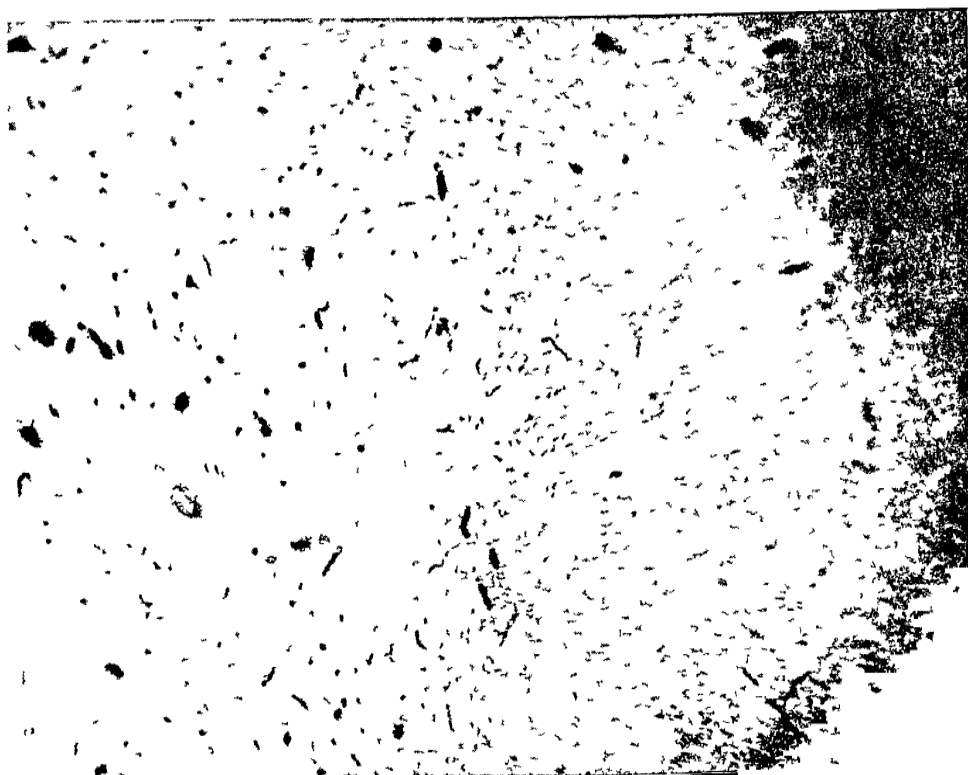


FIG 5 Multiple sclerosis Congestion and proliferation of vessels in a plaque (on left of photograph) in the cerebral white matter Masson stain, low power

Brown pigment was seen in endothelial and adventitial cells. Clots of fibrin and leukocytes occurred in vessels of various sizes, but they were less numerous and intense than in the cases of "encephalomyelitis". In some of them, invasion by endothelial cells could be demonstrated (figure 6). Complete occlusion of vessels by fibrous tissue was encountered with considerable frequency.

Can these changes be secondary to the parenchymal degeneration? A similar study was made of the vicinity of old softenings, which had produced massive secondary degenerations accompanied by dense gliosis. The vessels

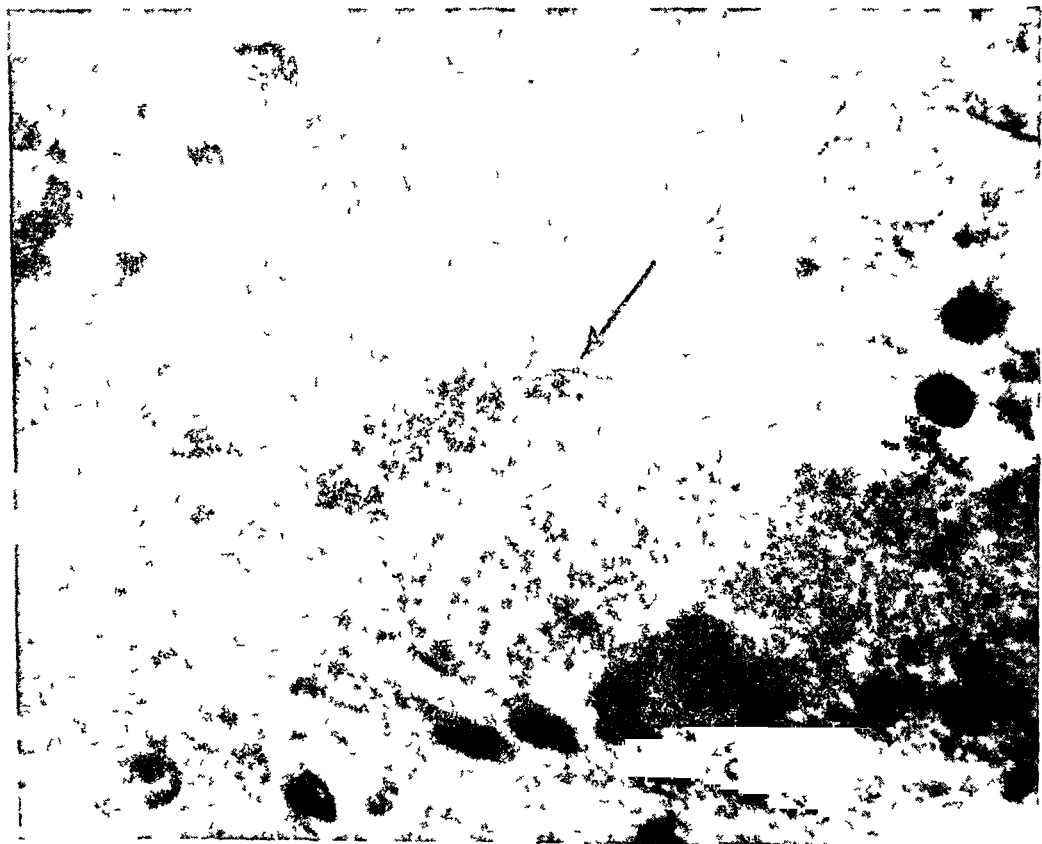


FIG 6 Multiple sclerosis. Platelet clot adherent to wall of vessel, with fibroblast (arrow) growing over the surface. Masson stain, oil immersion lens.

in such areas were approximately normal in size and number, not engorged, their walls contained no pigment, and their lumens contained no clots. There was often a thickening of the adventitia, which occasionally still held a few phagocytic cells.

Since, therefore, the vascular obstructions and dilatations seen in multiple sclerosis and "encephalomyelitis" are not regularly encountered in other types of degeneration, and since demyelination may be produced experimentally by venous obstructions, it seems reasonable to conclude that the

thromboses are primary to the changes in the nervous tissues, and not secondary. The histologic details will be discussed more fully in a paper now in preparation.

THE PART PLAYED BY TOXINS AND OTHER EXOGENOUS FACTORS

It has often been pointed out that the exacerbations of multiple sclerosis are apt to follow infections of various sorts. Pregnancy, trauma, chilling, the injection of foreign protein, and severe emotional strains are also mentioned as precipitating attacks. Von Hoesslin,²⁷ who has recently reviewed the subject, finds that one or the other of the factors mentioned appeared to be associated with the onset or progression of the disease in a large proportion of the cases which he studied. He concludes that there is probably also a predisposition to multiple sclerosis which is perhaps inherited.

If the lesions of multiple sclerosis are the result of thrombosis, it appears natural to seek the predisposing factor in the coagulating mechanism of the blood. Brickner²⁸ has described an abnormal lipolytic activity in the serum of patients suffering from multiple sclerosis, and McKenna²⁹ has pointed out that a leukocytosis is frequently found. It is not obvious, however, just how such abnormalities have a bearing on the presumed occurrence of thromboses.

The matter appears clearer since Solomon and Simon³⁰ demonstrated a functional abnormality in the clotting time in a series of patients with multiple sclerosis. Injection of adrenalin or of typhoid vaccine caused a more marked and more prolonged decrease in clotting time (that is, *increase* in coagulability) than in normal controls or patients with other neurologic diseases.

The mechanism of this increased lability of blood coagulation is still under investigation. It has been found that the blood fibrinogen is high in the majority of cases of multiple sclerosis examined. The blood calcium is within normal limits. No constant abnormality has been demonstrated in the platelet count, but in some instances there appears to be an abnormality in the behavior of the platelets. It is possible that some of the tests may be of diagnostic value.

The results of these investigations may be taken, then, to support the suggestion that some fundamental disorder of the blood leads it to clot in the cerebral venules of the central nervous system as a result of stimuli, infections, pregnancy or emotional upsets which normally would lead only to a mild increase in coagulability.

THERAPEUTIC IMPLICATIONS

No attempt has been made as yet to formulate a definite therapeutic program on the basis of the theory which has been outlined, but it is not difficult to visualize various methods which might be tried. It might be possible, for example, to lower the level of fibrinogen by diet, or to cause

a decrease in the number of platelets, or to administer a suitable anti-coagulant drug. It seems wiser, however, first to define the fundamental abnormality more clearly, if only because of the extraordinary difficulty of reaching a true estimate of the results of treatment simply from symptomatic and neurologic changes.

Obviously, the presumed exciting factors should be avoided if possible. Infections, accidents, and emotional upsets are not easily circumvented, but female patients should be advised not to become pregnant.

Certain negative conclusions are evident. Little is to be hoped for from antiseptic treatment. Heat treatment and cervical sympathectomy can have scarcely more than temporary value, if any. Treatment by injection of foreign protein is certainly to be avoided.

SUMMARY

1 The histologic features of the lesions of multiple sclerosis have been produced experimentally in animals in two ways: by the injection of minute doses of tetanus toxin, and by obstruction of cerebral venules.

2 Thrombi and obstructed vessels have been observed in the lesions of multiple sclerosis, and in those of "post-infectious encephalomyelitis," apparently an acute form of the same pathologic process.

3 It has been recognized, although rather vaguely, that there may be an association between the exacerbations of multiple sclerosis and infections, pregnancy and accidents.

4 Recent clinical investigations have shown that, although the clotting time is usually within normal limits in cases of multiple sclerosis, the coagulability of the blood is greatly increased by the administration of adrenalin or of typhoid vaccine, as compared with controls.

5 Therefore, there is probably a fundamental abnormality in the blood in multiple sclerosis, which interacts with endogenous or exogenous factors to cause thrombi in the cerebral venules.

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